

Summer Analytics 2026 – Week 2 Hackathon Report

E-Commerce Conversion Prediction Using Machine Learning

Problem Statement

The objective of this hackathon was to predict whether a user would convert (make a purchase) based on demographic information, browsing behavior, device characteristics, traffic source, and historical purchase activity. The target variable, **Converted**, is a binary classification label indicating whether a conversion occurred. Model performance was evaluated using the **F1 Score**, which balances precision and recall.

Dataset Overview

The dataset consisted of user-level observations containing features such as age, income, city tier, device type, traffic source, pages viewed, products viewed, time spent on site, previous purchases, discount exposure, browser version, and campaign information.

The training dataset contained **10,000 records** and **14 columns**. Missing values were present in the **Age**, **Income**, and **Time_On_Site** features. The target distribution was moderately imbalanced, with approximately 69% non-converted users and 31% converted users.

Data Preprocessing

To prepare the data for modeling, missing numerical values were handled using **median imputation**, while categorical variables were processed using **most-frequent value imputation**. The categorical features **Device_Type** and **Traffic_Source** were transformed using **One-Hot Encoding** to convert them into machine-readable numerical representations.

Feature Engineering

Additional features were created to capture user engagement and purchasing behavior more effectively:

- **Pages_per_Product** = $\text{Pages_Viewed} / (\text{Products_Viewed} + 1)$
- **Income_per_Age** = $\text{Income} / (\text{Age} + 1)$
- **Engagement** = $\text{Pages_Viewed} \times \text{Time_On_Site}$

These engineered features were intended to provide richer information about browsing patterns and user engagement.

Model Development

An **XGBoost Classifier** was used as the primary machine learning model due to its strong performance on structured tabular datasets. A preprocessing pipeline was created to combine imputation, encoding, and model training into a single reproducible workflow.

The model was initially trained using the training dataset and validated using the provided **public_test.csv** dataset. Since the competition metric was F1 Score, probability threshold tuning was performed to identify the threshold that maximized validation performance.

Validation Results

The optimal classification threshold identified during validation was:

- **Best Threshold:** 0.27
- **Best F1 Score:** 0.5391

This threshold produced better performance than the default threshold of 0.50 and was therefore used for final prediction generation.

Final Approach

After validation, the training and public validation datasets were combined to maximize the amount of labeled data available for learning. The final XGBoost model was retrained on the combined dataset and used to generate predictions for **private_test.csv**. These predictions were exported in the required submission format as **submission.csv**.

Conclusion

The final solution employed data preprocessing, feature engineering, threshold optimization, and an XGBoost classification model to predict user conversions. The approach leveraged both the training and public validation datasets to improve generalization and generate predictions for the hidden evaluation dataset.